

Kalp Shah

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EDUCATION

Indian Institute of Technology, Gandhinagar

Gandhinagar, India

Bachelor of Technology in Electrical Engineering, Minor in Computer Science and Engineering July 2024 – April 2028

EXPERIENCE

Research Intern

May 2026 – July 2026

Machine Intelligence and Resilience Lab, IIT Gandhinagar

Gujarat, India

- Engineered a Computer Vision pipeline to estimate Flood Depth from CCTV Footage using Deep Learning Models
- Curated a Street View Flood Image Dataset with 1000+ Images, utilizing ResNet18 for Data Filtering
- Trained YOLO Models to classify Flood Depth Levels based on Cars as reference objects

Software Development Intern

May 2025 – July 2025

AB6 Robotics

Gujarat, India

- Built a cross-platform desktop application in Python and Qt enabling real-time communication between robots and students across 15+ robotics challenges, actively used by 30+ students
- Integrated a RAG-based conversational assistant into the platform to query robotics concepts

PROJECTS

Landing Location for Mars Rover using Classical Computer Vision

Supervisor: Prof. Shanmuganthan Raman — Github — Research Poster

Jan. 2026 – Apr. 2026

- Developed a computer vision pipeline to identify safe landing zones on Mars under computational constraints
- Designed a modular system utilizing Sobel derivatives for slope detection, Fast Fourier Transforms for texture analysis, and the Maximal Rectangle Algorithm for zone identification
- Achieved a verifiable, low-power solution with less than 2 seconds runtime on consumer-level hardware

Real-time Carbon Credits Market Analysis using Agentic AI

Inter-IIT Tech Meet 14.0 — Github

Dec 2025

- Engineered a realtime financial streaming engine using Pathway, Kafka and Debezium (CDC), capturing live PostgreSQL database changes to reduce data freshness latency from 60 minutes to < 3s
- Architected a multi-agent RAG system leveraging LLMs and hybrid inference to deliver explainable, real-time investment insights
- Proposed the “Green Investment Index” (GII) metric, a Quantitative Alpha Signal computed via incremental stateful aggregations, delivered to clients through WebSocket API

CPU Ray Tracing Engine in C++

Personal Project — Github

- Developed a physics based ray tracing engine in C++ to simulate photorealistic rendered scenes
- Implemented features like metallic reflections, dielectric refraction, shadows, and anti-aliasing
- Optimized rendering performance using mathematical vector operations and parallel process handling to decrease frame render times from 2 to 0.09 seconds (22x speedup)

Limit Orderbook and Trade Matching Engine

Personal Project — Github

- Built a high-performance Limit Order Book and trade matching engine in C++ with support for multiple order types including Market, Limit, Fill-or-Kill (FOK), and Immediate-or-Cancel (IOC) orders
- Implemented low-latency price-time priority matching to support 4000 Transactions per second
- Applied systems programming concepts including concurrency-aware design, cache-efficient structures, and deterministic event processing

Liquid Volume Measurement System using YF-S201 Flow Sensor

Supervisor: Prof. Arup Lal Chakraborty (ES116) — Project Report

- Delivered a portable solution to measure liquid volumes with a constrained cost of Rs. 1000
- Interfaced a YF-S201 flow sensor with an Arduino microcontroller to process the signals generated
- Achieved high accuracy in volume measurement by testing known volumes to calculate and apply a specific calibration factor adjusted for fluid viscosity

TECHNICAL SKILLS

Languages: Python, C/C++, SQL (MySQL), JavaScript, HTML/CSS, Verilog, Assembly(MIPS)

Frameworks and Tools: React, Next.js, Node.js, Flask, Git Docker

Libraries: PyTorch, OpenCV, scikit-learn, Langchain, Pandas, Numpy, Matplotlib

Skills: Data Structures and Algorithms, Machine Learning, System Design, Web Development, Problem Solving

ACHIEVEMENTS

- Expert on Codeforces (Handle: kalpshah1808, peak rating: 1720) with a country rank of 650 out of 98000
- Ranked 47th amongst 2,500+ Participants at The Integral Cup 2026 Round 1
- Ranked 1359 amongst 13,000+ Participants at IICPC Codefest Prelims
- Awarded a Branch Change to Electrical Engineering based on CGPA after the first year
- Secured AIR 6637 in JEE Advanced and AIR 3917 in JEE Mains amongst 1.4 Million Candidates
- Selected as an Academic Discussions Hour Mentor for ES115 Course(Design, Innovation and Prototyping)
- Selected for Inter-IIT Soccer Team

EXTRACURRICULAR ACTIVITIES

Technical Council, IIT Gandhinagar

Technical Coordinator

May 2026 – Present

- Developed Technical Event Proposals and coordinated with administration
- Overseeing 10+ Clubs with 500+ Members cumulatively

Outreach Coordinator

May 2025 - Apr. 2026

- Led outreach initiatives for hackathon partnerships with colleges and corporate sponsorships
- Reactivated Social Media Presence on Instagram(tech.iitgn) and LinkedIn(tech-council-iitgn)

Entrepreneurship Initiative, IIT Gandhinagar

Dec. 2024 – Apr. 2025

Technical Team

- Worked on the EII Team Website (eii-iitgn.github.io)

Amalthea, Annual Technical Summit of IIT Gandhinagar

Aug. 2024 – Jan. 2025

Junior Sponsorship Executive

- Developed Sponsorship Proposals for collaboration with companies and getting funding
- Conducted meetings and secured funding from major corporations
- Learnt teamwork, leadership, and organizational skills

RELEVANT COURSEWORK

Computer Science and Engineering: Computing, Data-centric Computing, Data Structures and Algorithms, Introduction to Competitive Programming, Artificial Intelligence, Digital Systems, Machine Learning[†], Computer Vision[†]
Mathematics: Probability, Statistics and Data Visualization, Calculus of Single Variable and Linear Algebra, Ordinary Differential Equations, Calculus of Several Variables, Numerical Methods

[†] - currently ongoing course